Business Understanding Report

Loan Eligibility Preprocessing & Feature Engineering

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A. Business Problem

Financial institutions receive a large volume of loan applications and must decide quickly whether to approve or reject each request. Manual review is time-consuming, inconsistent, and costly. Incorrect approvals increase credit risk and potential losses, while incorrect rejections reduce revenue and damage customer relationships. The organisation therefore needs a reliable, data-driven method to assess loan eligibility based on applicant characteristics such as income, credit history, education, employment status, and loan details.

The business problem is therefore to understand the characteristics associated with successful loan applications and prepare the available applicant data for predictive modelling.

The project seeks to answer the following question: **Can historical applicant and loan information be transformed into a reliable dataset that can be used to predict whether a future loan application is likely to be approved?**

B. Project Objective

The objective of this week's work is to transform the raw Loan Prediction dataset into a clean, well structured, machine-learning-ready format. This involves handling missing values, encoding categorical variables, scaling numerical features, detecting and treating outliers, removing redundant columns, and performing basic feature selection.

The prepared dataset will serve as the foundation for training classification models in Week 3 that predict whether a loan application should be approved (Yes/No).

C. Target Variable

The target variable is **Loan_Status** (categorical: Y / N).

- **Y** – Loan application was approved
- **N** – Loan application was rejected

Because the target contains two possible outcomes, the future machine learning learning problem will be treated as a **binary classification problem.** The training set contains approximately 614 labelled applications; the test set contains unlabelled applications for which the model will generate predictions.

The dataset also includes predictor variables covering applicant demographics (Gender, Married, Dependents, Education, Self_Employed), financial details (ApplicantIncome, CoapplicantIncome, LoanAmount, Loan_Amount_Term), credit history (Credit_History), and property location (Property_Area), all of which are candidate features for predicting the target.

D. Importance of Predictive Modelling

Predictive modelling enables the organisation to:

- Automate the initial screening of loan applications and reduce manual processing time.
- Apply consistent, evidence-based decision criteria across all applications.
- Identify the applicant characteristics most strongly associated with approval or rejection (especially Credit_History).
- Support risk management by highlighting high-risk profiles before funds are disbursed.

E. Expected Business Impact

- **Faster decisions:** Automating an initial risk assessment reduces the time to a loan decision for straightforward applications.
- **Consistency:** Applicants with similar profiles are assessed against the same criteria, reducing reviewer-to-reviewer variability.
- **Efficient use of credit officer time:** Officers can focus on borderline or high-risk applications flagged by the model rather than every application equally.
- **Better risk visibility:** A documented, feature-based view of what drives approval decisions gives the lender clearer insight into its own risk exposure.
- **A foundation for scale:** A clean, well-engineered dataset and pipeline can be extended as application volumes grow, without re-doing this preparation work from scratch.